

HDOC DAY – 23

ALGORITHM:

Step 1: Start.

Step 2: Declare choice as an integer.

Step 3: Display the menu:

Step 4: Read the user's choice.

Step 5: Use a switch statement to perform the selected operation.

Step 6: For each choice:

Case 1 – Solid Square

- Call pattern1().
- Input number of rows n.
- Use an outer loop from 1 to n for rows.
- Use an inner loop from 1 to n for columns.
- Print * in every position.

Case 2 – Solid Rectangle

- Call pattern2().
- Input rows m and columns n.
- Outer loop runs m times.
- Inner loop runs n times.
- Print * in every position.

Case 3 – Row-Number Pattern

- Call pattern3().
- Input rows m and columns n.
- Use nested loops.
- Print the value of the row number i in every column.

Case 4 – Column-Number Pattern

- Call pattern4().
- Input rows m and columns n.
- Use nested loops.
- Print the value of the column number j in every row.

Case 5 – Exit

- Exit the program.

Default

- Display "Invalid choice!".

Step 7: Repeat the menu until the user selects Exit.

Step 8: Stop.

C PROGRAM:

```
#include <stdio.h>
```

```
/* Function declarations */
```

```
void pattern1(void);
```

```
void pattern2(void);
```

```
void pattern3(void);
```

```
void pattern4(void);
```

```
int main()
```

```
{
```

```
    int choice;
```

```
    do
```

```
    {
```

```
        printf("\n--- Basic Square/Rectangular Patterns Menu ---\n");
```

```
        printf("1. Solid square pattern\n");
```

```
        printf("2. Solid rectangular pattern\n");
```

```
printf("3. Row-number pattern\n");  
printf("4. Column-number pattern\n");  
printf("5. Exit\n");
```

```
printf("Enter your choice: ");  
scanf("%d", &choice);
```

```
switch(choice)
```

```
{
```

```
    case 1:
```

```
        pattern1();
```

```
        break;
```

```
    case 2:
```

```
        pattern2();
```

```
        break;
```

```
    case 3:
```

```
        pattern3();
```

```
        break;
```

```
    case 4:
```

```
        pattern4();
```

```
        break;
```

```
    case 5:
```

```
        printf("Exiting program...\n");
```

```
        break;
```

```
        default:

            printf("Invalid choice!\n");

        }

    } while(choice != 5);

    return 0;
}


/* 1. Solid square pattern */
void pattern1(void)
{
    int n, i, j;

    printf("Enter the number of rows: ");
    scanf("%d", &n);

    if(n <= 0)
    {
        printf("Number of rows must be positive.\n");
        return;
    }

    for(i = 1; i <= n; i++)
    {
        for(j = 1; j <= n; j++)
```

```
    {  
        printf("* ");  
    }  
    printf("\n");  
}  
}
```

/* 2. Solid rectangular pattern */

void pattern2(void)

```
{  
    int m, n, i, j;  
  
    printf("Enter the number of rows: ");  
    scanf("%d", &m);  
  
    printf("Enter the number of columns: ");  
    scanf("%d", &n);  
  
    if(m <= 0 || n <= 0)  
    {  
        printf("Rows and columns must be positive.\n");  
        return;  
    }  
  
    for(i = 1; i <= m; i++)  
    {  
        for(j = 1; j <= n; j++)
```

```
{  
    printf("* ");  
}  
printf("\n");  
}  
}
```

/* 3. Row-number pattern */

void pattern3(void)

```
{  
    int m, n, i, j;  
  
    printf("Enter the number of rows: ");  
    scanf("%d", &m);  
  
    printf("Enter the number of columns: ");  
    scanf("%d", &n);  
  
    if(m <= 0 || n <= 0)  
    {  
        printf("Rows and columns must be positive.\n");  
        return;  
    }  
  
    for(i = 1; i <= m; i++)  
    {  
        for(j = 1; j <= n; j++)
```

```
    {  
        printf("%d ", i);  
    }  
    printf("\n");  
}  
}
```

/* 4. Column-number pattern */

```
void pattern4(void)
```

```
{  
    int m, n, i, j;  
  
    printf("Enter the number of rows: ");  
    scanf("%d", &m);  
  
    printf("Enter the number of columns: ");  
    scanf("%d", &n);
```

```
    if(m <= 0 || n <= 0)  
    {  
        printf("Rows and columns must be positive.\n");  
        return;  
    }
```

```
    for(i = 1; i <= m; i++)  
    {  
        for(j = 1; j <= n; j++)
```

```

{
    printf("%d ", j);

}

printf("\n");

}

}

```

TEST CASE TABLE:

TEST CASE	CHOICE	INPUT	EXPECTED RESULT
1	1	n = 4	Prints a 4 × 4 solid square
2	2	m = 3, n = 5	Prints a 3 × 5 solid rectangle
3	3	m = 4, n = 4	Each row contains its row number
4	4	m = 4, n = 4	Each column contains its column number
5	1	n = 0	Displays error message
6	2	m = -2, n = 5	Displays error message
7	7	--	Displays "Invalid choice!"
8	5	--	Exits the program

```
— Basic Square/Rectangular Patterns Menu —
1. Solid square pattern
2. Solid rectangular pattern
3. Row-number pattern
4. Column-number pattern
5. Exit
Enter your choice: 1
Enter the number of rows: 4
* * * *
* * * *
* * * *
* * * *

— Basic Square/Rectangular Patterns Menu —
1. Solid square pattern
2. Solid rectangular pattern
3. Row-number pattern
4. Column-number pattern
5. Exit
Enter your choice: 2
Enter the number of rows: 3
Enter the number of columns: 5
* * * * *
* * * * *
* * * * *

— Basic Square/Rectangular Patterns Menu —
1. Solid square pattern
2. Solid rectangular pattern
3. Row-number pattern
4. Column-number pattern
5. Exit
Enter your choice: 4
Enter the number of rows: 4
Enter the number of columns: 4
1 2 3 4
1 2 3 4
1 2 3 4
1 2 3 4

— Basic Square/Rectangular Patterns Menu —
1. Solid square pattern
2. Solid rectangular pattern
3. Row-number pattern
4. Column-number pattern
5. Exit
Enter your choice: █
```